

Trainings ▾

Preparatory Steps

with Mechanically Actuated Injector

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.



⚠ WARNING ⚠

Do not remove the pressure cap from a hot engine. Wait until the temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant or steam can cause personal injury.

⚠ WARNING ⚠

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

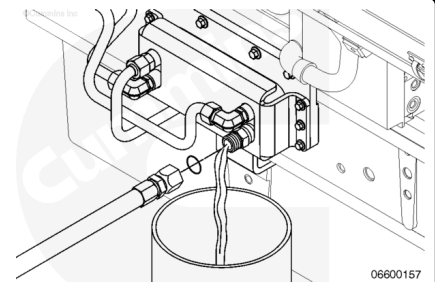
- Disconnect the battery cables. Refer to Procedure 013-009 (</qs3/pubsys2/xml/en/procedures/56/56-013-009.html>).
- Disconnect the air supply line. Refer to the OEM Service Manual.
- Drain the cooling system. Refer to Procedure 008-018 (</qs3/pubsys2/xml/en/procedures/56/56-008-018-tr.html>).

Remove

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⚠ WARNING ⚠

Depending on the circumstances, fuel is flammable. Keep all cigarettes, flames, pilot lights, arching equipment, and switches out of the work area and areas sharing ventilation to reduce the possibility of severe personal injury or death when working on the fuel system.



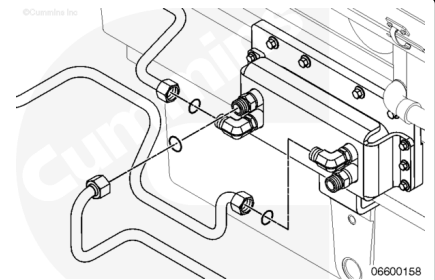
Remove the fuel drain line that connects the fuel cooler to the fuel tank. Drain the fuel from the fuel cooler into a container.

Disconnect the fuel drain to fuel tank line from the fuel cooler.

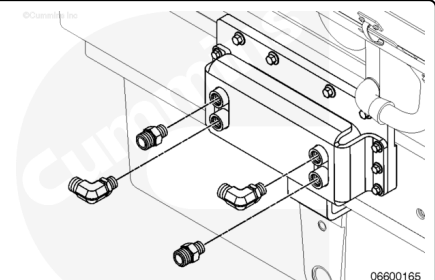
Remove and discard the o-ring.

Disconnect the coolant supply and return lines from the fuel cooler.

Remove and discard the o-rings.



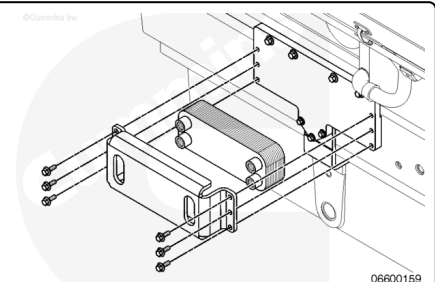
Remove the connectors from the fuel cooler fuel and coolant ports.



The fuel cooler is positioned between the front and rear cooler brackets, and is **not** directly connected with mounting capscrews.

Remove the six front bracket mounting capscrews.

Remove the front fuel cooler bracket and the fuel cooler.



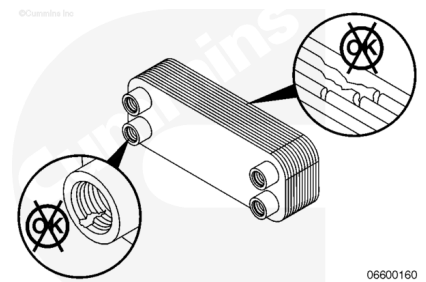
Inspect for Reuse

with Mechanically Actuated Injector

Inspect the fuel cooler plates for damage. If the fuel cooler plates or shell is damaged the fuel cooler **must** be replaced.

Inspect the fuel cooler fuel and coolant port threads for damage.

If the threads are damaged beyond repair the fuel cooler **must** be replaced.



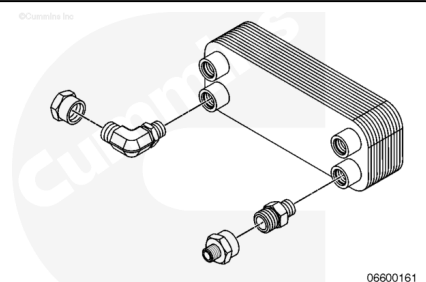
Pressure Test

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Install the connectors into the fuel cooler fuel ports.

Place a 7/8-14 UN-28 threaded cap on one fuel connector.

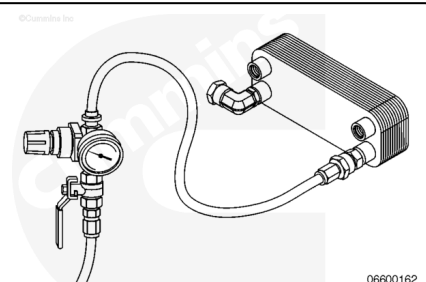
Place a 7/8-14 UN-28 threaded cap fitted with an air fitting on the other connector.



Install a pressure regulator onto the cap and air fitting.

Connect shop air to the pressure regulator.

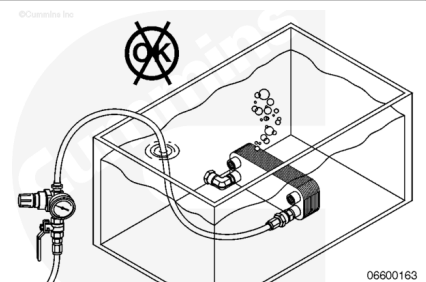
Adjust the pressure regulator to 170 kPa [25 psi].



⚠ CAUTION ⚠

To reduce the possibility of damage to the fuel cooler, the water in the test tank must be treated with a noncorrosive additive.

Submerge the fuel cooler in a tank of treated water for 10 minutes.

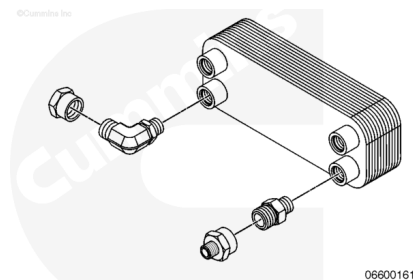


If air bubbles escape from the fuel side of the fuel cooler in the last two minutes of the test, the fuel cooler **must** be replaced.

Remove the fuel cooler from the tank of treated water.

Remove the pressure regulator from the air fitting.

Remove the caps from the fuel connectors of the fuel cooler.

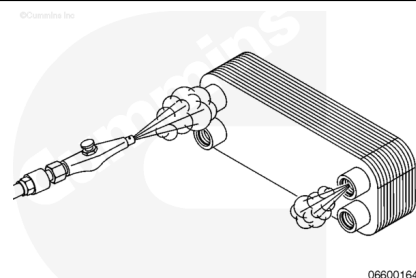


⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying dirt and debris can cause personal injury.

Drain the treated water out of the fuel cooler.

Use compressed air to dry the coolant side of the fuel cooler.



Install

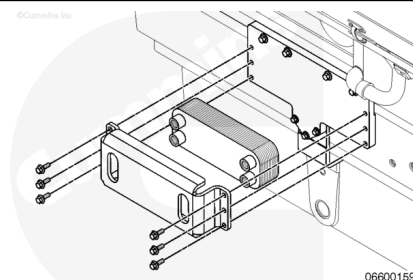
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Place the fuel cooler and fuel cooler front bracket onto the engine.

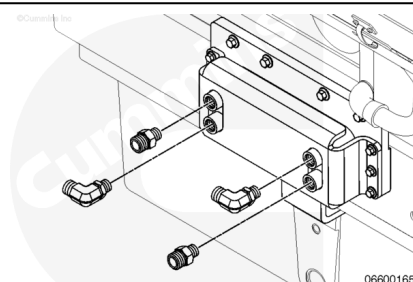
Thread the six fuel cooler front bracket capscrews into the fuel cooler rear bracket.

Tighten the capscrews.

Torque Value: 45 n•m [33 ft-lb]



Install the connectors into the fuel and coolant ports of the fuel cooler.

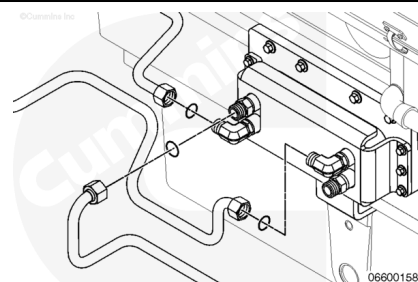


Install the o-rings onto the fuel and coolant connectors.

Install the fuel and coolant lines.

Tighten the fuel and coolant lines.

Torque Value: 27 n•m [20 ft-lb]

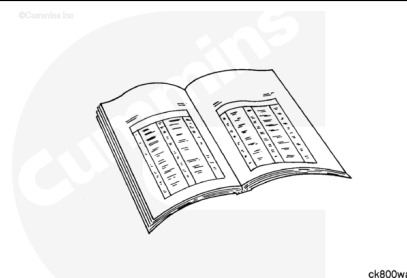


Finishing Steps

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- Connect the battery cables. Refer to Procedure 013-009 (</qs3/pubsys2/xml/en/procedures/56/56-013-009.html>).
- Connect the air supply line. Refer to the OEM Service Manual.
- Fill the cooling system. Refer to Procedure 008-018 (</qs3/pubsys2/xml/en/procedures/56/56-008-018-tr.html>).